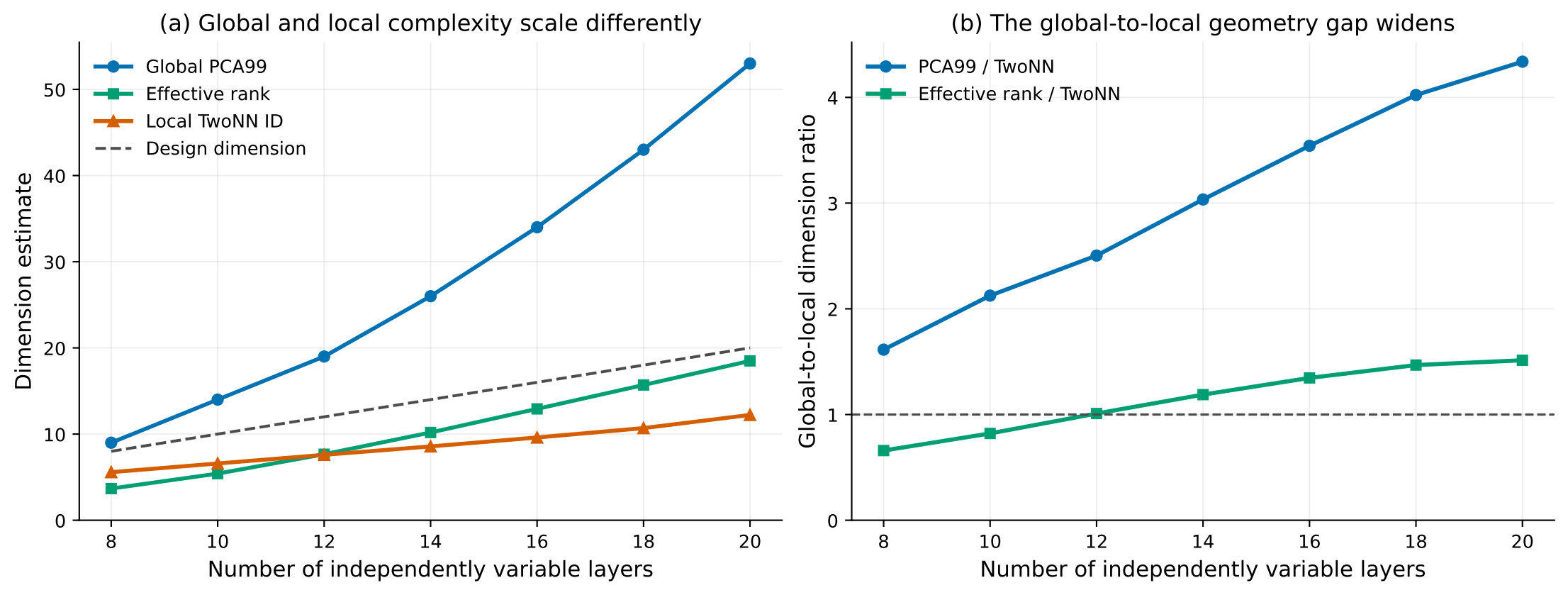
(a) Global and local complexity scale differently
Global PCA99
Effective rank
Local TwoNN ID
Design dimension
Dimension estimate
0
10
20
30
40
50
8
10
12
14
16
18
20
Number of independently variable layers
(b) The global-to-local geometry gap widens
PCA99 / TwoNN
Effective rank / TwoNN
Global-to-local dimension ratio
0
1
2
3
4
8
10
12
14
16
18
20
Number of independently variable layers